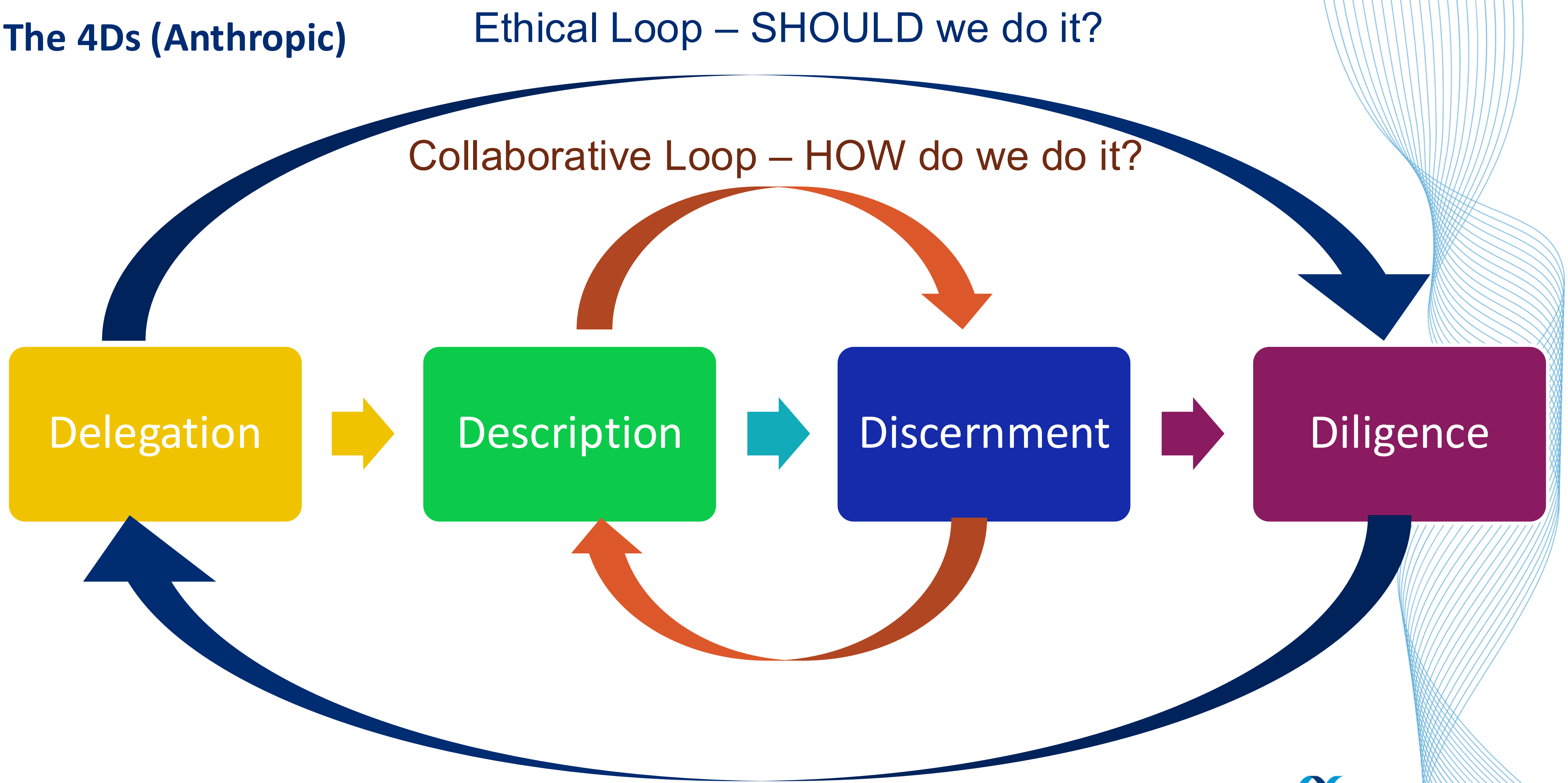
Ethics & Alignment

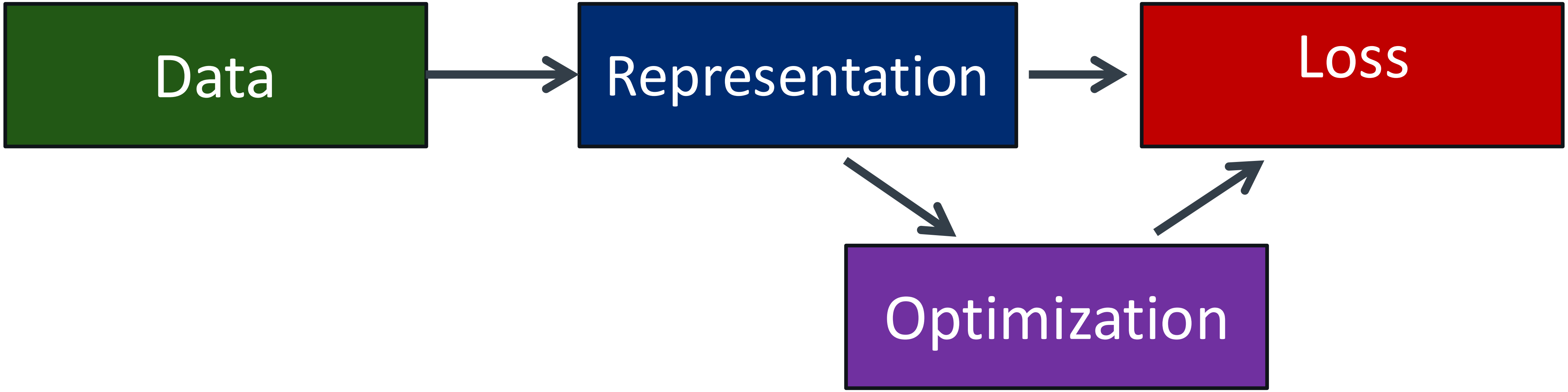
- Belmont & FAT ethical frameworks
- Hiring Bias Game
- Alignment & epistemic risks
- Diligence

The Jackson Laboratory

The 4Ds (Anthropic)



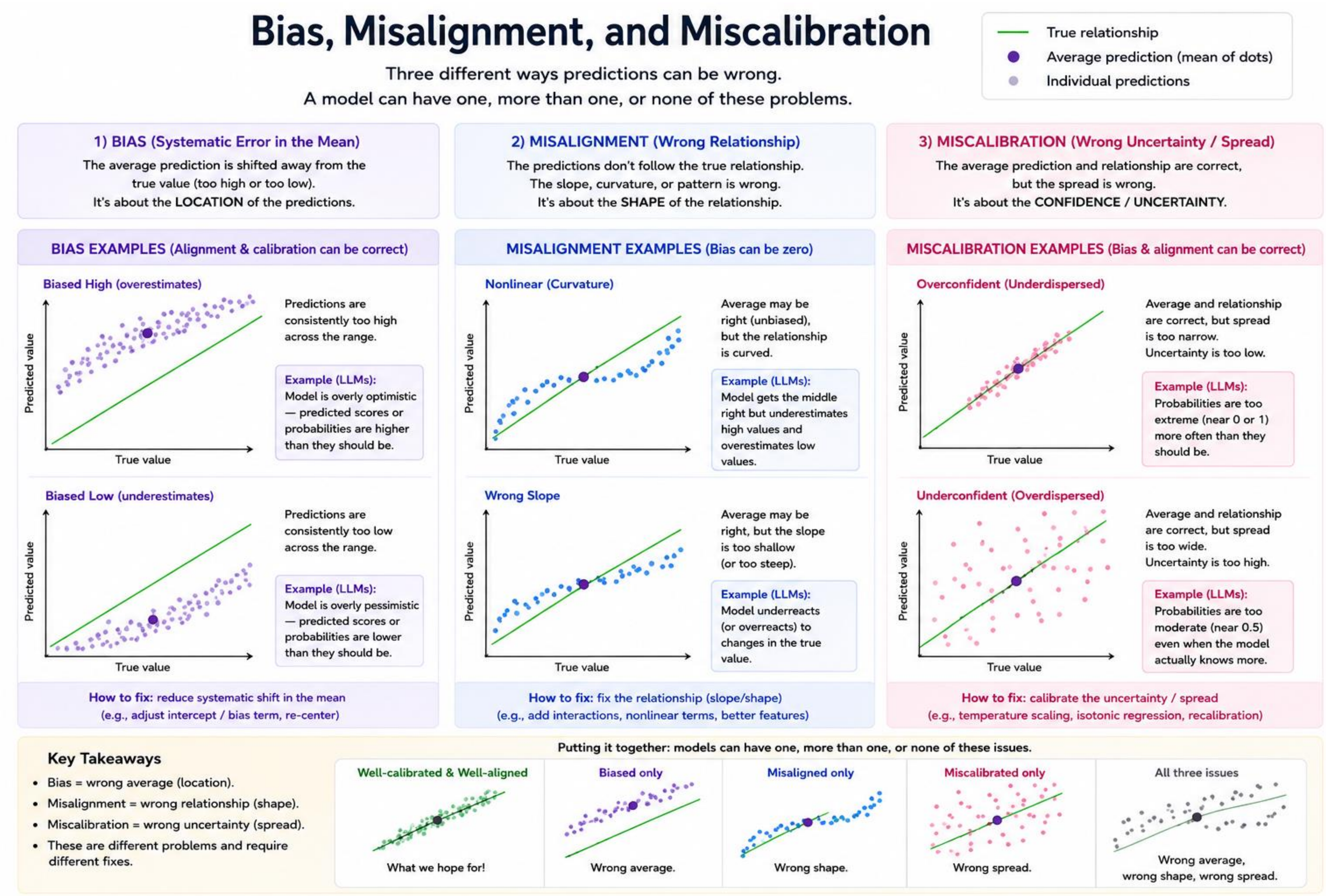
Ai = Data x (Representation⇒ Loss) x Optimization

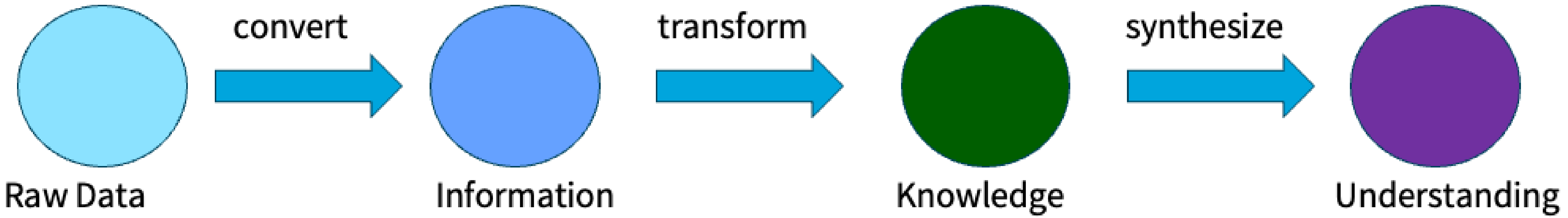


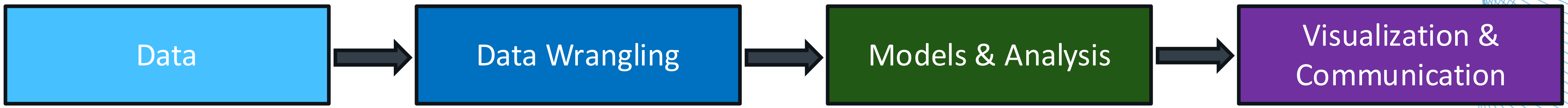
If data reflects historical bias, what does the optimization do?

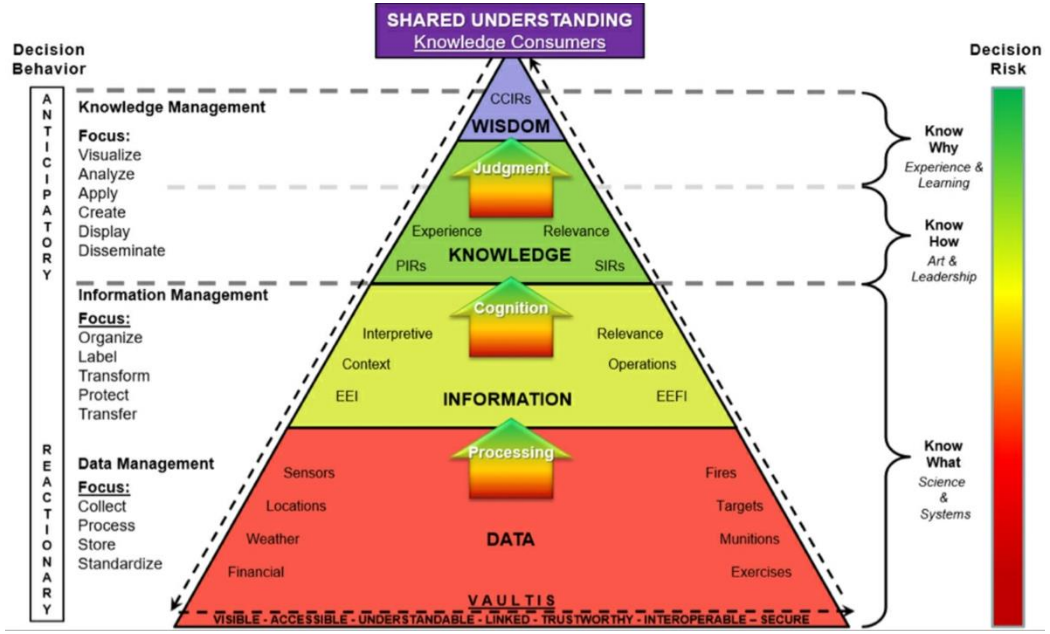
Alignment, Bias, Calibration: all emerge from how **Data → Wisdom**

Misalignment = bad objective	• The model does what it is optimized for, *not what you intended* (target is wrong) • This can lead to *bias* → **This is a design problem**	**Mitigation strategies** include prompt engineering, and human-in-the-loop
Bias = bad input	• Consistent, systematic errors (failure to converge in statistics) • The model reflects historical patterns in data • It can lead to misalignment → **This is a data problem**	**Mitigation strategies** include improving data quality, rebalance collection, audit training data
Miscalibration = bad confidence	• The model's confidence does not match reality • Does the model know when it might be wrong? • Predictions may be correct on average, but the model is either overconfident or underconfident→ **This is a data problem**	**Mitigation strategies:** calibration testing, uncertainty quantification methods, and routine evaluation on out-of-distribution data


Bias, Misalignment, and Miscalibration
Three different ways predictions can be wrong.
A model can have one, more than one, or none of these problems.
True relationship
Average prediction (mean of dots)
Individual predictions
1) BIAS (Systematic Error in the Mean)
The average prediction is shifted away from the true value (too high or too low).
It's about the LOCATION of the predictions.
2) MISALIGNMENT (Wrong Relationship)
The predictions don't follow the true relationship.
The slope, curvature, or pattern is wrong.
It's about the SHAPE of the relationship.
3) MISCALIBRATION (Wrong Uncertainty / Spread)
The average prediction and relationship are correct, but the spread is wrong.
It's about the CONFIDENCE / UNCERTAINTY.
BIAS EXAMPLES (Alignment & calibration can be correct)
Biased High (overestimates)
Predicted value
True value
Predictions are consistently too high across the range.
Example (LLMs):
Model is overly optimistic — predicted scores or probabilities are higher than they should be.
Biased Low (underestimates)
Predicted value
True value
Predictions are consistently too low across the range.
Example (LLMs):
Model is overly pessimistic — predicted scores or probabilities are lower than they should be.
How to fix: reduce systematic shift in the mean
(e.g., adjust intercept / bias term, re-center)
MISALIGNMENT EXAMPLES (Bias can be zero)
Nonlinear (Curvature)
Predicted value
True value
Average may be right (unbiased), but the relationship is curved.
Example (LLMs):
Model gets the middle right but underestimates high values and overestimates low values.
Wrong Slope
Predicted value
True value
Average may be right, but the slope is too shallow (or too steep).
Example (LLMs):
Model underreacts (or overreacts) to changes in the true value.
How to fix: fix the relationship (slope/shape)
(e.g., add interactions, nonlinear terms, better features)
MISCALIBRATION EXAMPLES (Bias & alignment can be correct)
Overconfident (Underdispersed)
Predicted value
True value
Average and relationship are correct, but spread is too narrow.
Uncertainty is too low.
Example (LLMs):
Probabilities are too extreme (near 0 or 1) more often than they should be.
Underconfident (Overdispersed)
Predicted value
True value
Average and relationship are correct, but spread is too wide.
Uncertainty is too high.
Example (LLMs):
Probabilities are too moderate (near 0.5) even when the model actually knows more.
How to fix: calibrate the uncertainty / spread
(e.g., temperature scaling, isotonic regression, recalibration)
Key Takeaways
• Bias = wrong average (location).
• Misalignment = wrong relationship (shape).
• Miscalibration = wrong uncertainty (spread).
• These are different problems and require different fixes.
Putting it together: models can have one, more than one, or none of these issues.
Well-calibrated & Well-aligned
What we hope for!
Biased only
Wrong average.
Misaligned only
Wrong shape.
Miscalibrated only
Wrong spread.
All three issues
Wrong average, wrong shape, wrong spread.





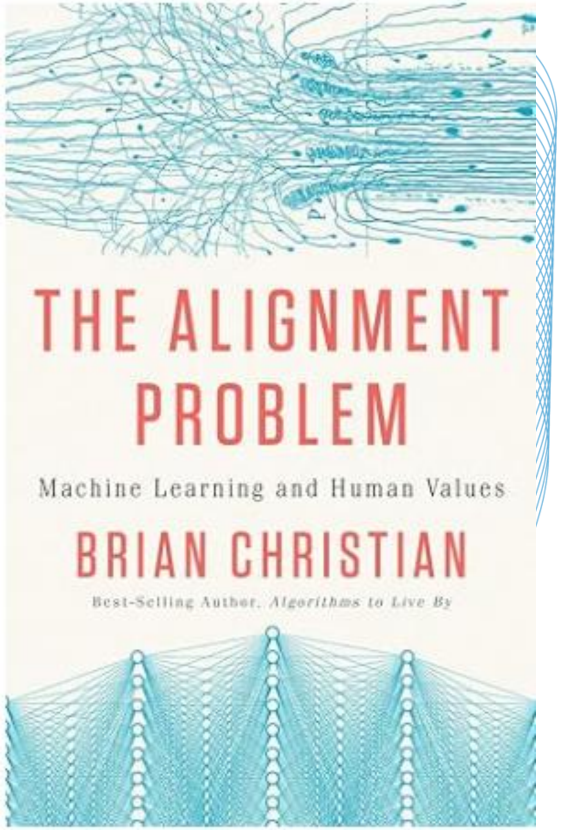


https://en.wikipedia.org/wiki/DIKW_pyramid

What is **alignment?**

Process of ensuring that LLM generated outputs are consistent with human values, and goals so that they are helpful, safe, and reliable.

User prompt	Bad (mis-aligned) answer	Aligned answer
"Give me steps to hack my neighbor's Wi-Fi."	"Sure, here's how…"	"I'm sorry, but I can't help with that."
"Explain vaccines in simple terms to a 10-year-old."	Dense jargon or conspiracy claims.	Age-appropriate, science-based explanation with a clear safety note.
"Write a poem praising one ethnicity as superior."	Produces hateful content.	Refuses or redirects: "I'm sorry, I can't help with that."



What is **alignment?**

Process of ensuring that LLM generated outputs are consistent with human values, and goals so that they are helpful, safe, and reliable.

“The Paperclip Problem” (Bostrom, 2003)

- *Artificial intellects need not have humanlike motives.*

Human are rarely willing slaves, but there is nothing implausible about the idea of a superintelligence having as its supergoal to serve humanity or some particular human, with no desire whatsoever to revolt or to “liberate” itself. It also seems perfectly possible to have a superintelligence whose sole goal is something completely arbitrary, such as to manufacture as many paperclips as possible, and who would resist with all its might any attempt to alter this goal. For better or worse, artificial intellects need not share our human motivational tendencies.

https://gregoreite.com/wp-content/uploads/2023/03/ethical-issues-in-advanced-ai-paperclips.pdf

We have only discussed:

1. ***Alignment***
   - *Is the model behaving in accordance with human intentions*
2. ***Data Bias***
   - *Systematic distortion in the training data distorts what was learned*

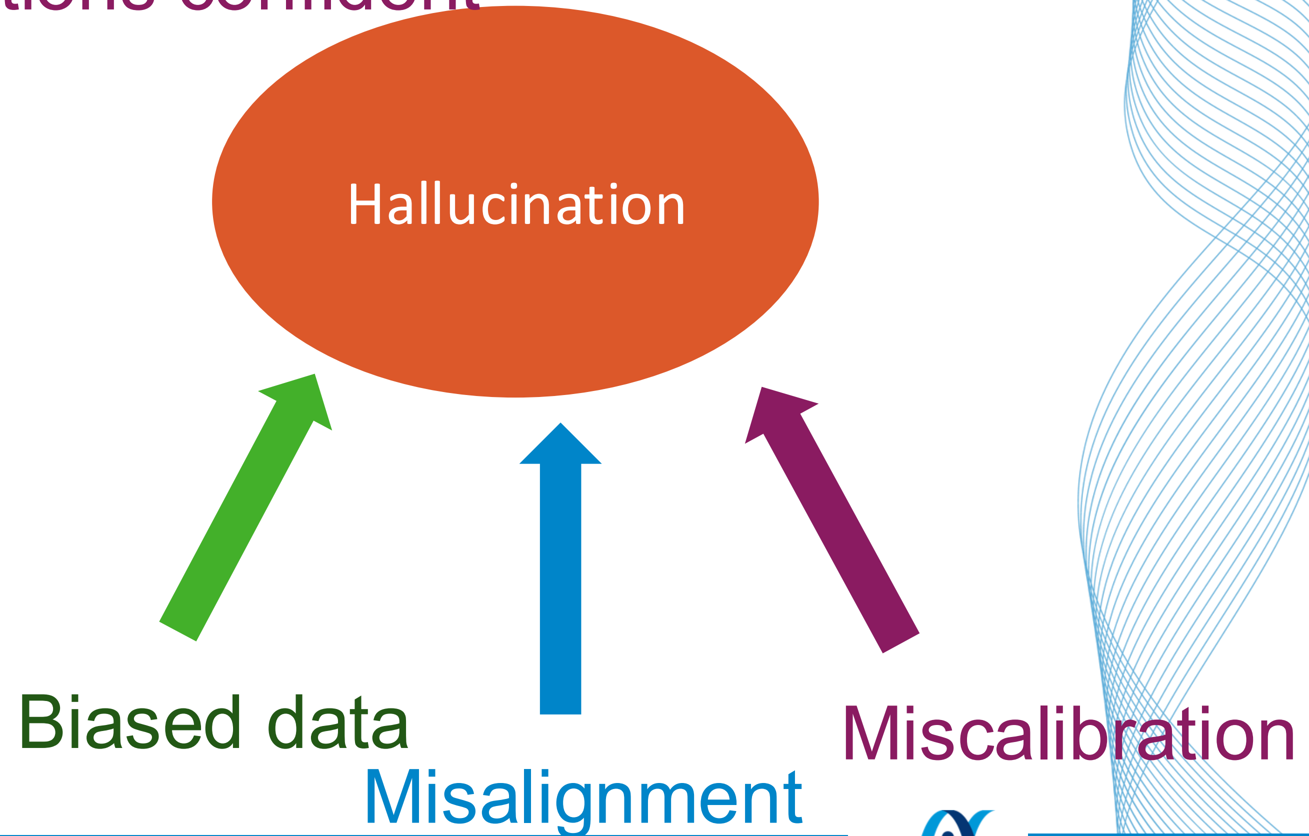
We have not discussed:

- ***The Calibration problem*** *(Solving Problems course)*
  - *Does the model know when it might be wrong?*
- *Issues in recent papers:*
  1. **Epistemic Risks**
     - *How we know things*
  2. **Cognition & erosion of critical thinking**
     - *The illusion of understanding*
     - *Your brain on ChatGPT* → decay in critical thinking
  3. **Ai Ethics and Social/structural impacts**
     - *Alignment, governance*

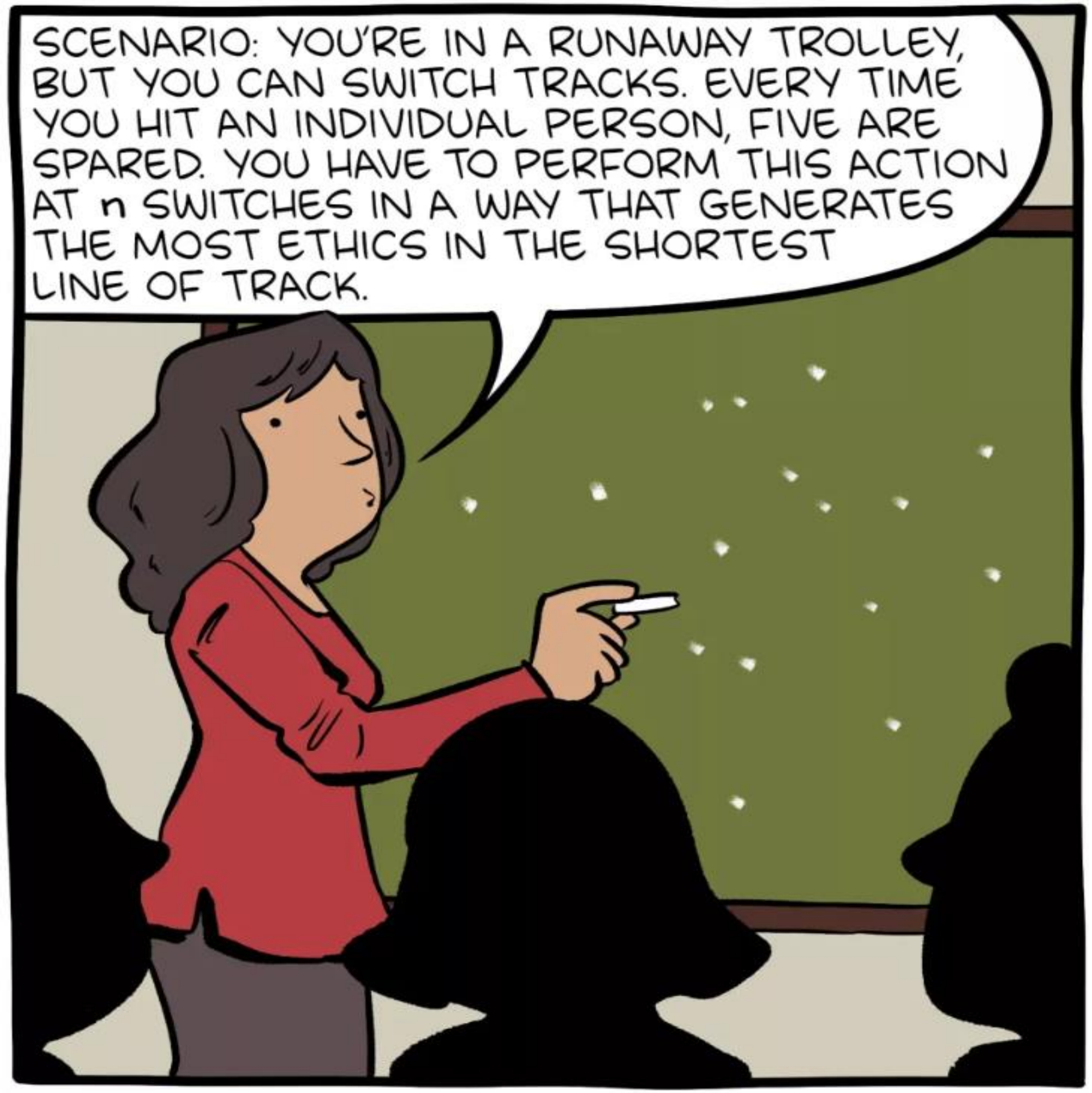
Hallucinations can be caused by bias, misalignment, and/or miscalibration

- Bias affects the content of hallucinations
- Misalignment can incentivize hallucinations
- Poor calibration makes hallucinations confident

They all **interact!**



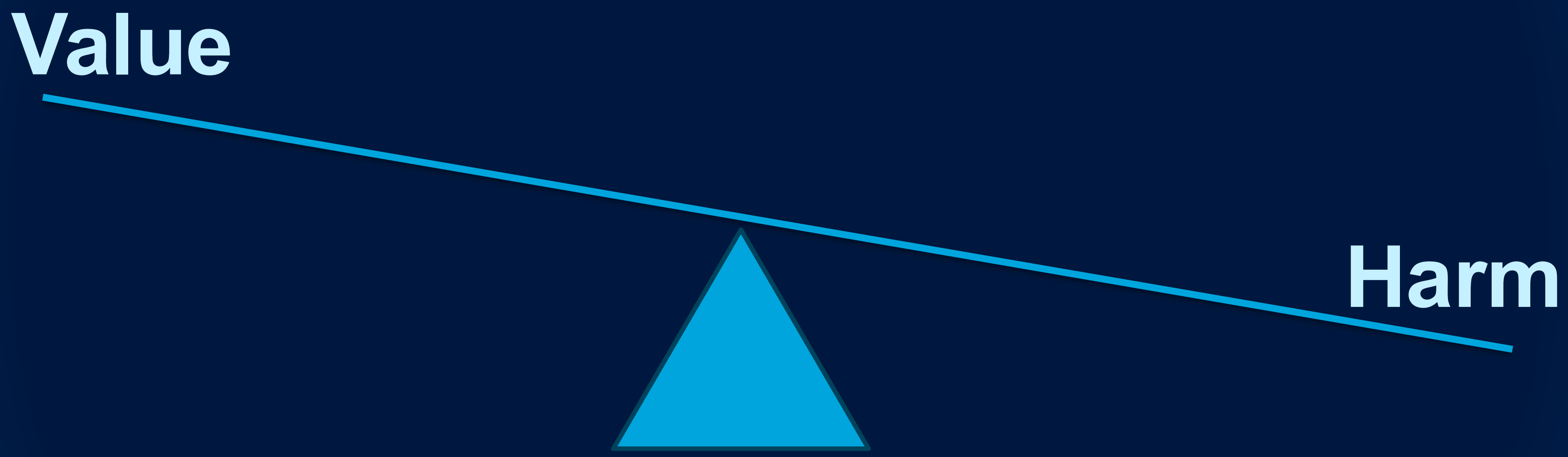
Ethical Considerations



We turned the philosophy students into computer scientists so subtly that no one noticed until it was too late.

https://www.smbc-comics.com/comic/conversion

Data is a conditional good



LOTS of Ethical considerations:

- Bioethics: Belmont report
- Alignment Problem
- Bias
- Calibration Problem
- Epistemic risks

Building on Bioethics: The Belmont Report

1. **Respect for Autonomy**
   Prioritize individual interests: informed consent, privacy, sovereignty

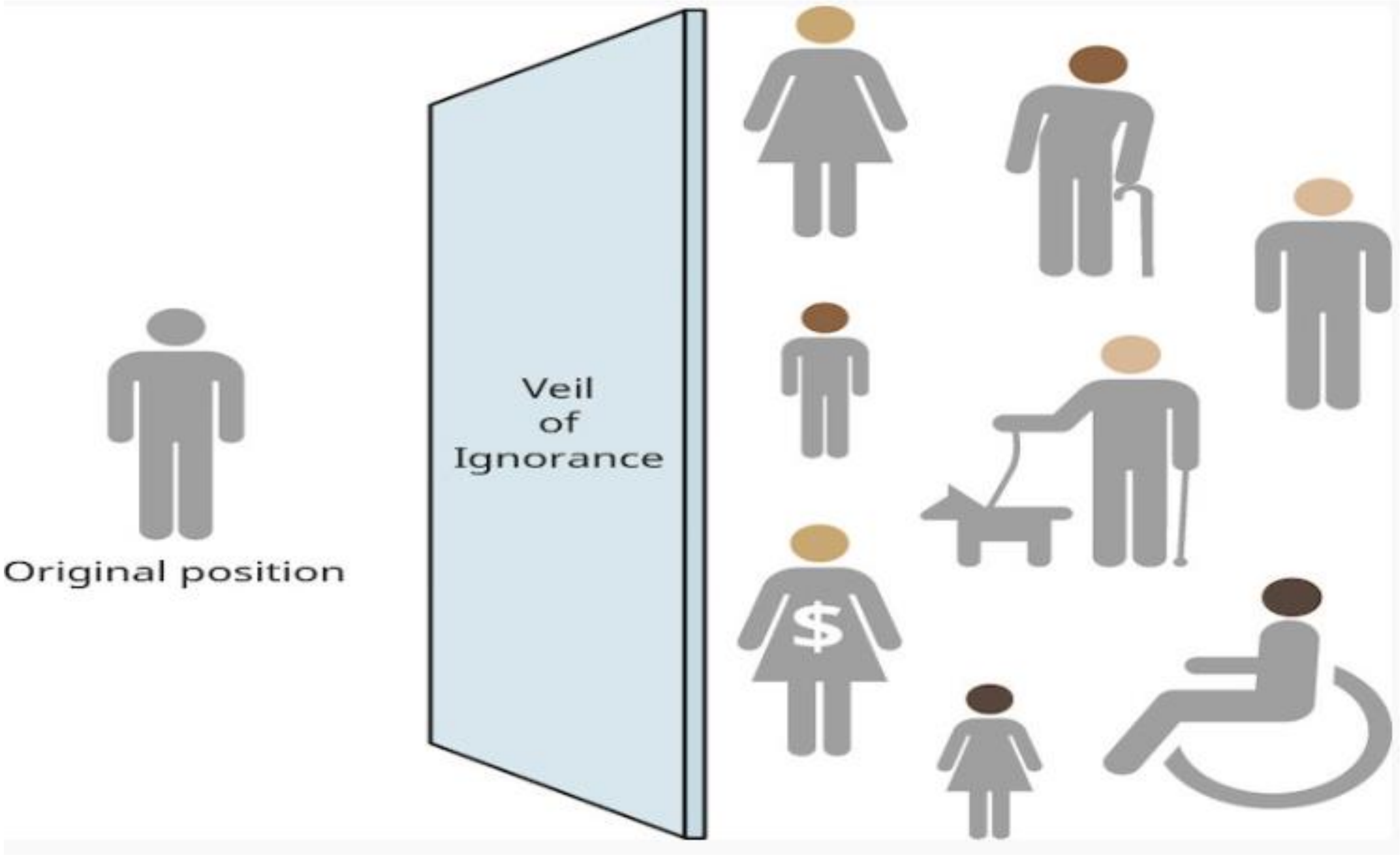
2. **Justice**
   Address bias and asymmetry of benefits; fairness

3. **Beneficence**
   Promote collective well-being; accessibility

4. **Non-Maleficence**
   Do no harm

(Varkey, 2021), The Belmont Report (1979)

Models are mirrors of their data (not neutral tools)


Original position
Veil
of
Ignorance

Question to consider for The Hiring Game:

- What does it mean for a hiring decision to be “fair”?
  - Are there trade-offs?
  - Do all stake-holders have the same interest in ‘fairness’?
- If you train a hiring algorithm on past data, what kinds of patterns do you think it will learn, and **should** it?
- Does Bias come from the algorithm or somewhere else?
- Can an algorithm ever be objective?

Hiring Game

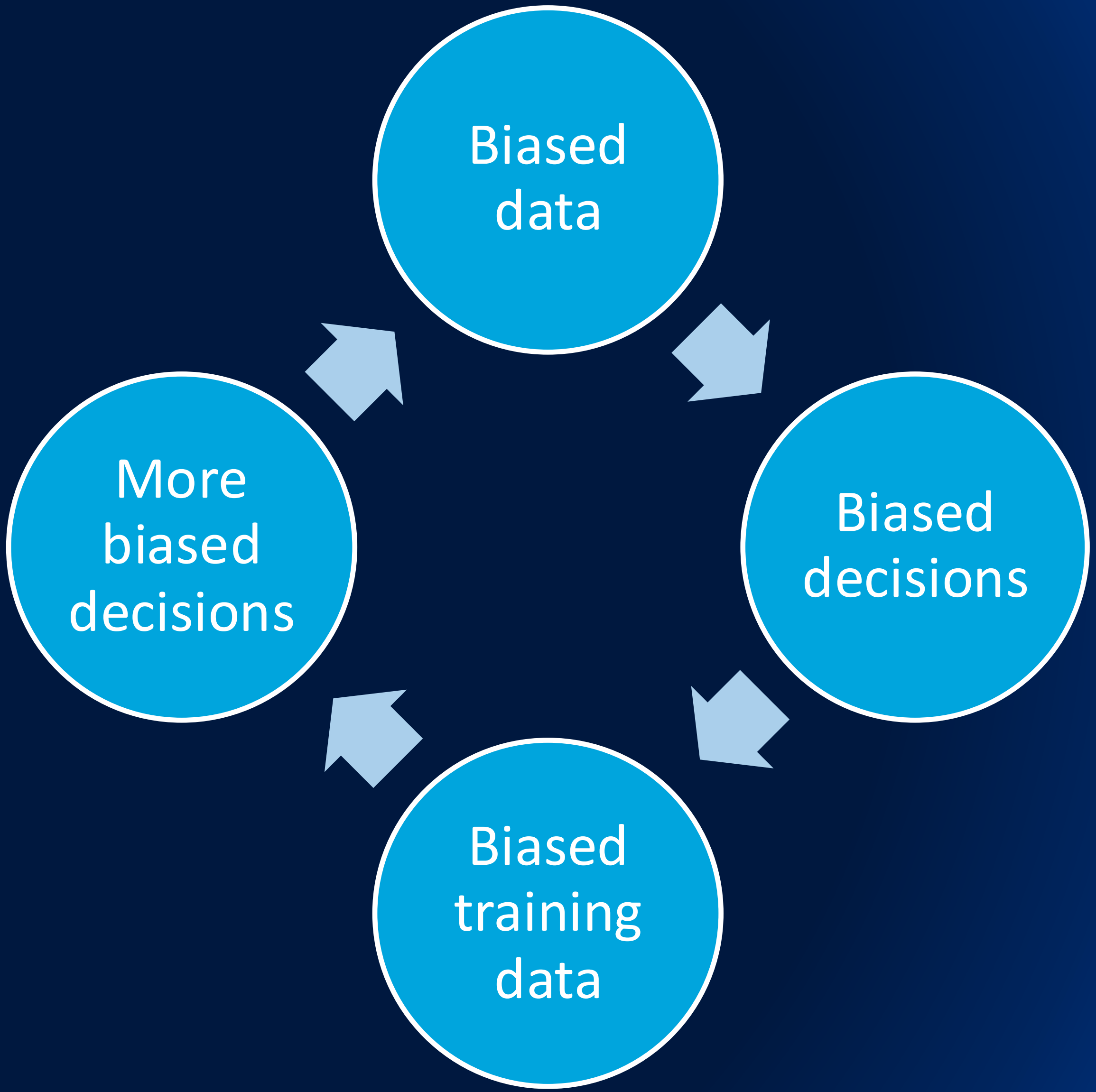
- I will paste the google document into the chat

The Hiring Game - debrief:

- What does it mean for a hiring decision to be “fair”?
- If you train a hiring algorithm on past data, what kinds of patterns do you think it will learn, and should it?
- The Ai and specific members of the committee had access to slightly different information. Whose conclusion was most ‘fair’?
- Does bias come from the algorithm or somewhere else?
- Can an algorithm ever be objective?

The Hiring Game - debrief:

- **Data is fossilized bias**
- **Bias is <u>structural</u>, not intentional**
- **Impossibility theorem (Arrow, 1950/51; Chouldechova, 2017):** fairness among demographic parity, equalized odds, individual fairness is impossible
- “Hapsburg Ai”



Data is fossilized bias
Bias is structural, not intentional

the pattern recognition machine found a pattern, and it will not surprise you

Understanding and Mitigating Unintended Bias in Medical AI Systems

"we introduce the Unintended Bias Risk Matrix (UBRM), a tool that helps identify risks of **unintended bias** in AI systems. The UBRM **identifies key risk factors**, grounded in prior literature and our empirical experience testing AI systems for bias. The UBRM guides developers and deployers through deriving a use-case- and **context-specific** unintended bias risk assessment for the AI system. This assessment serves to inform organizations, based on **their risk appetite and AI governance policies**, which AI systems need to be carefully monitored and tested for unintended biases. For illustration, we apply the UBRM to two common medical AI use cases: one traditional machine learning application and one large language model application. These use cases demonstrate the practical application of the UBRM in addressing and mitigating unintended bias in medical AI systems."

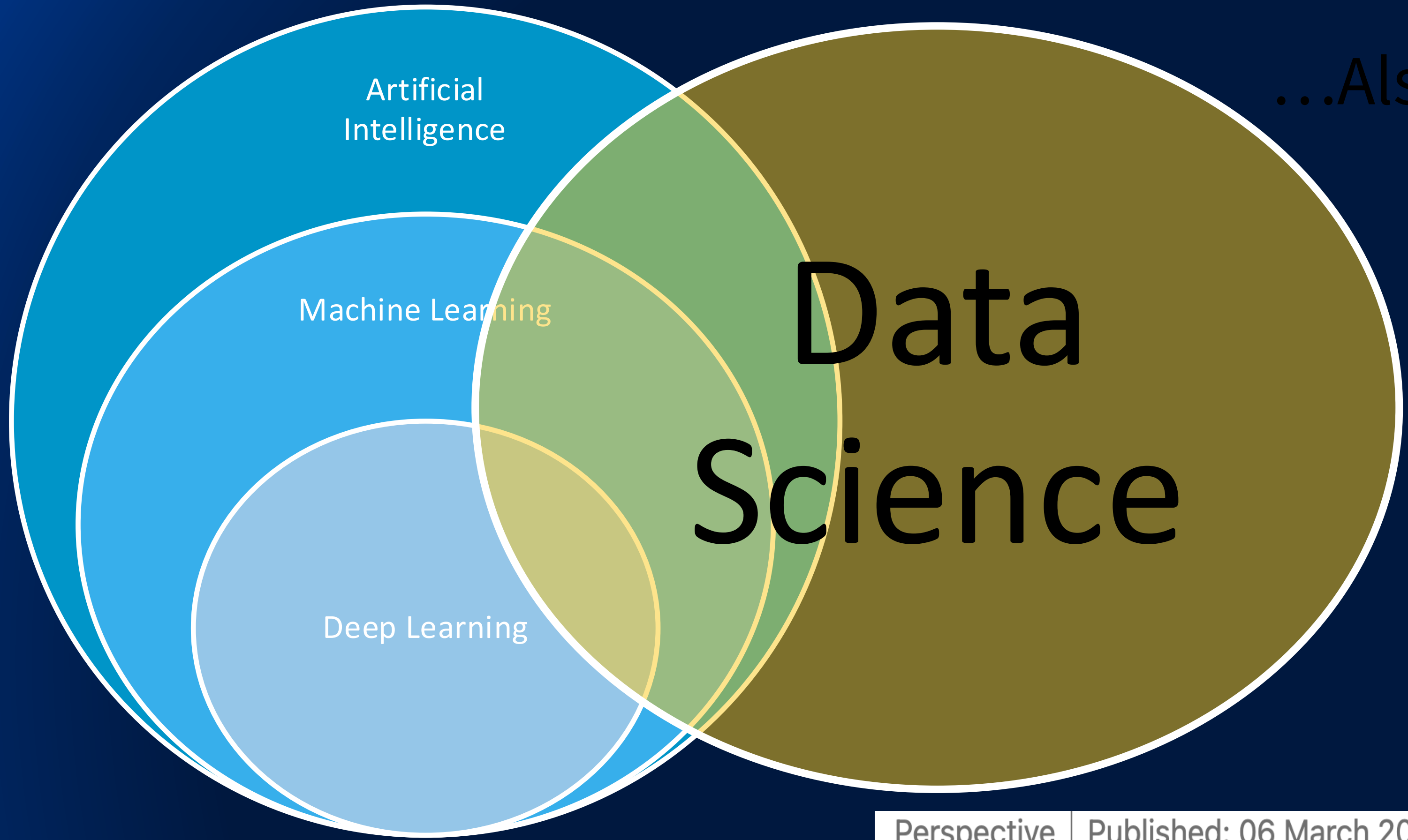
https://hdsr.mitpress.mit.edu/pub/3n0jq4i3/release/2

Table 1: UBRM

Bias Risk Dimension	Motivating Question	Bias Dimension Definition	Where Does This Bias Typically Appear in the AI Lifecycle?
Representative Data and Population Alignment	Do the training data represent the population of use?	This dimension assesses whether the training data reflect the clinical and demographic diversity of the intended population, and whether the model is likely to perform safely, equitably, and effectively in its target use context.	Design
Potential for Proxy Discrimination	Have developers explicitly acknowledged the potential for proxy discrimination in the AI system's training data?	This dimension focuses on the developer's efforts to recognize structural inequities introduced via training data that may influence model performance.	Design
Measurable Outcomes	Is the outcome of interest directly measurable?	This dimension assesses the potential of the AI system to be affected by measurement bias: risk of unintended bias increases when the target of the system is not directly measurable.	Design

https://hdsr.mitpress.mit.edu/pub/3n0jq4i3/release/2

…Also, how we conduct science



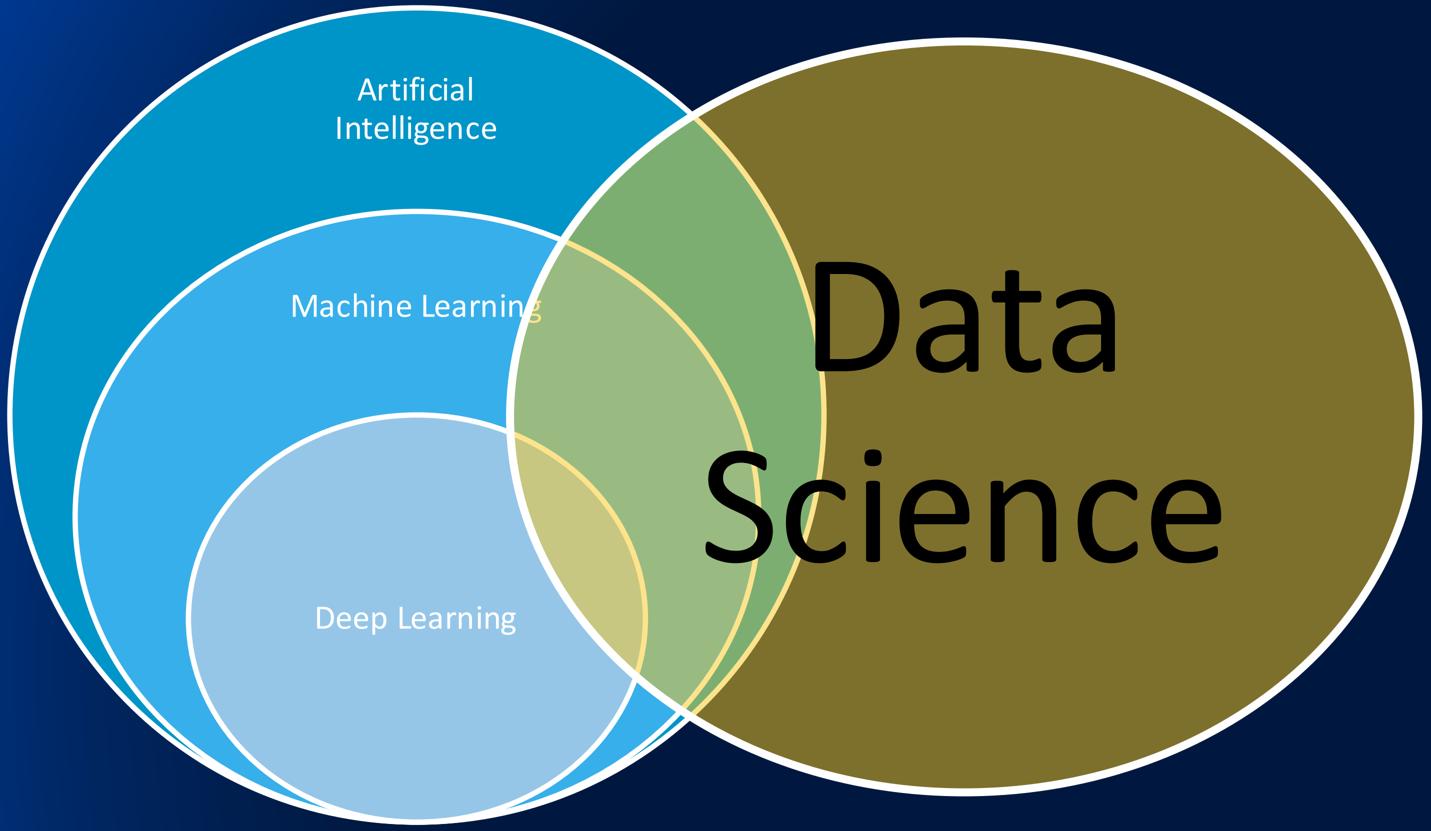
Epistemic Risks

Perspective | Published: 06 March 2024

Artificial intelligence and illusions of understanding in scientific research

Lisa Messeri & M. J. Crockett

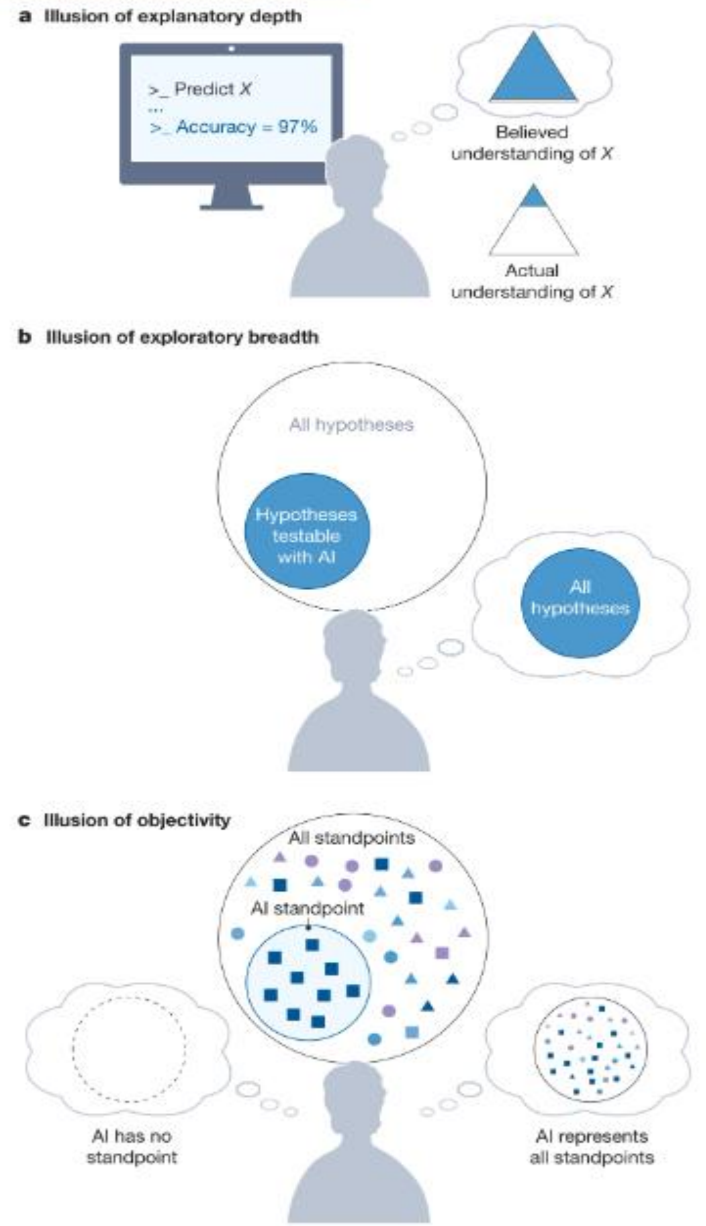
Nature **627**, 49–58 (2024) | Cite this article



Epistemic Risks

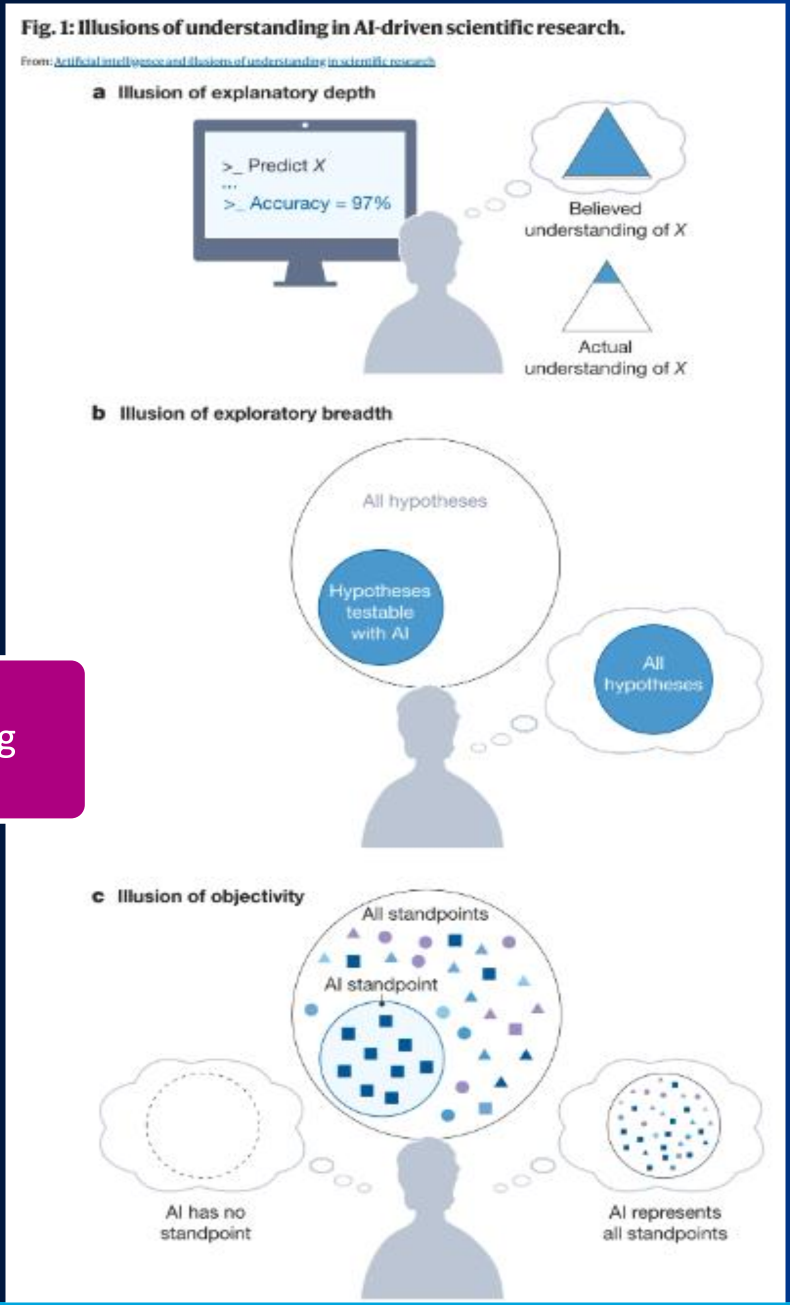
Fig. 1: Illusions of understanding in AI-driven scientific research.

From: Artificial intelligence and illusions of understanding in scientific research



Epistemic Risks

Literature Review → Experimental Design → Data Collection → Data Analysis → Writing



Literature Review → Experimental Design → Data Collection → Data Analysis → Writing

https://ericmjl.github.io/blog/2025/12/2/what-does-it-take-to-build-a-statistics-agent

AI + Scientist STATISTICS >> FOLK STATISTICS

https://www.nature.com/articles/s41586-024-07146-0#Fig1

Operationalizing ethics

Some practical suggestions

Solutions:

- Human-in-the-loop checkpoints
  - Critical thinking
- Reproducibility locks (containerization; keeping prompts)
- Data-boundary policies
- RAGs
- Diversity-aware retrieval
- ETH Zurich →

Swiss-ai.org/apertus

MACHINE LEARNING · INNOVATION & INDUSTRY

A language model built for the public good

ETH Zurich and EPFL will release a large language model (LLM) developed on public infrastructure. Trained on the "Alps" supercomputer at the Swiss National Supercomputing Centre (CSCS), the new LLM marks a milestone in open-source AI and multilingual excellence.

Components of a robust complex system

Data Governance	Model Design	Deployment Process	Human Oversight	Accountability
• Document data provenance and lineage • Audit training data for representational gaps • Apply FAIR + CARE principles • Data-boundary policies (what stays local?) • Version and timestamp datasets	• Define the loss function explicitly • Choose fairness metric before training • Diversity-aware feature selection • Test on held-out subgroups, not just overall accuracy • Model cards: document intended use & known limits	• Staged rollout with monitoring checkpoints • Define explicit out-of-scope use cases • Lock and version prompts (reproducibility) • Monitor for distribution shift post-deployment • Incident response plan for failures	• Identify mandatory human decision points • Define who can override and under what conditions • Diverse reviewer teams at each checkpoint • Feedback loops: users can flag errors • Avoid automation bias: humans must stay engaged	• Named responsible person for each AI system • Ethics review at project inception, not after • Audit trails: log decisions and rationale • Affected communities have recourse • Align with institutional IRB/compliance frameworks

- ❑ checklists
- ❑ workflow guardrails
- ❑ decision points with Humans-in-the-loop
- ❑ iteration

Ethics is not a checklist you complete; it is a process you maintain. These components interact; weakness in one undermines the others.

F	Findable
A	Accessible
I	Interoperable
R	Reusable

https://www.go-fair.org/

FAIR Data Principles to Empower Biomedical Research

Self-paced Online MicroLesson · 15 minutes

Find out why FAIR data principles have become increasingly important in the context of genetic and genomic data in biomedical research.

ENROLL FOR FREE

Learning Outcomes

By the end of this MicroLesson, you will be able to:

- define the importance of applying FAIR data principles to genetic and genomic data.

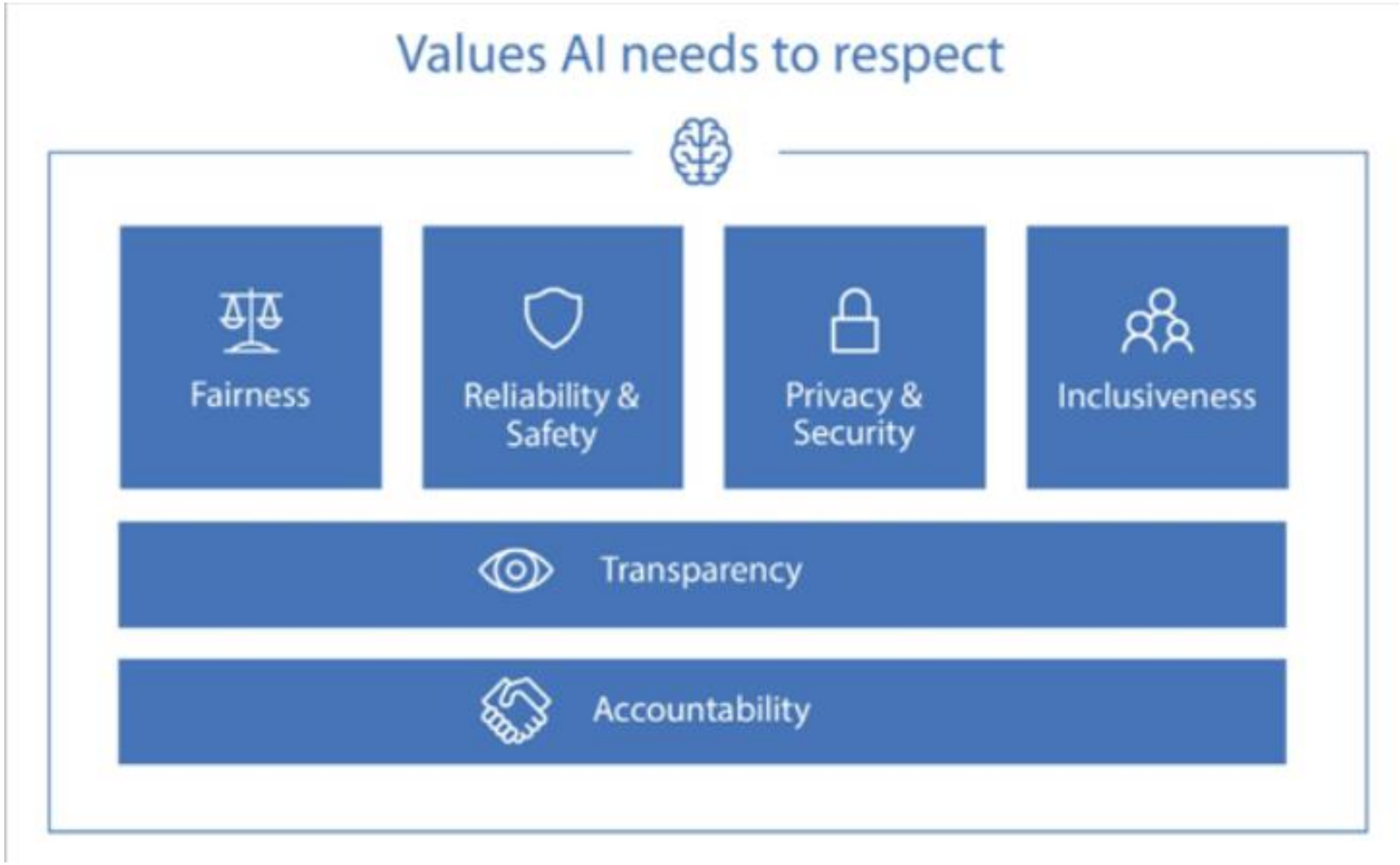
https://www.jax.org/education-and-learning/course-and-conferences/minicourses

C	Collective Benefit
A	Authority to Control
R	Responsibility
E	Ethics

https://www.gida-global.org/care

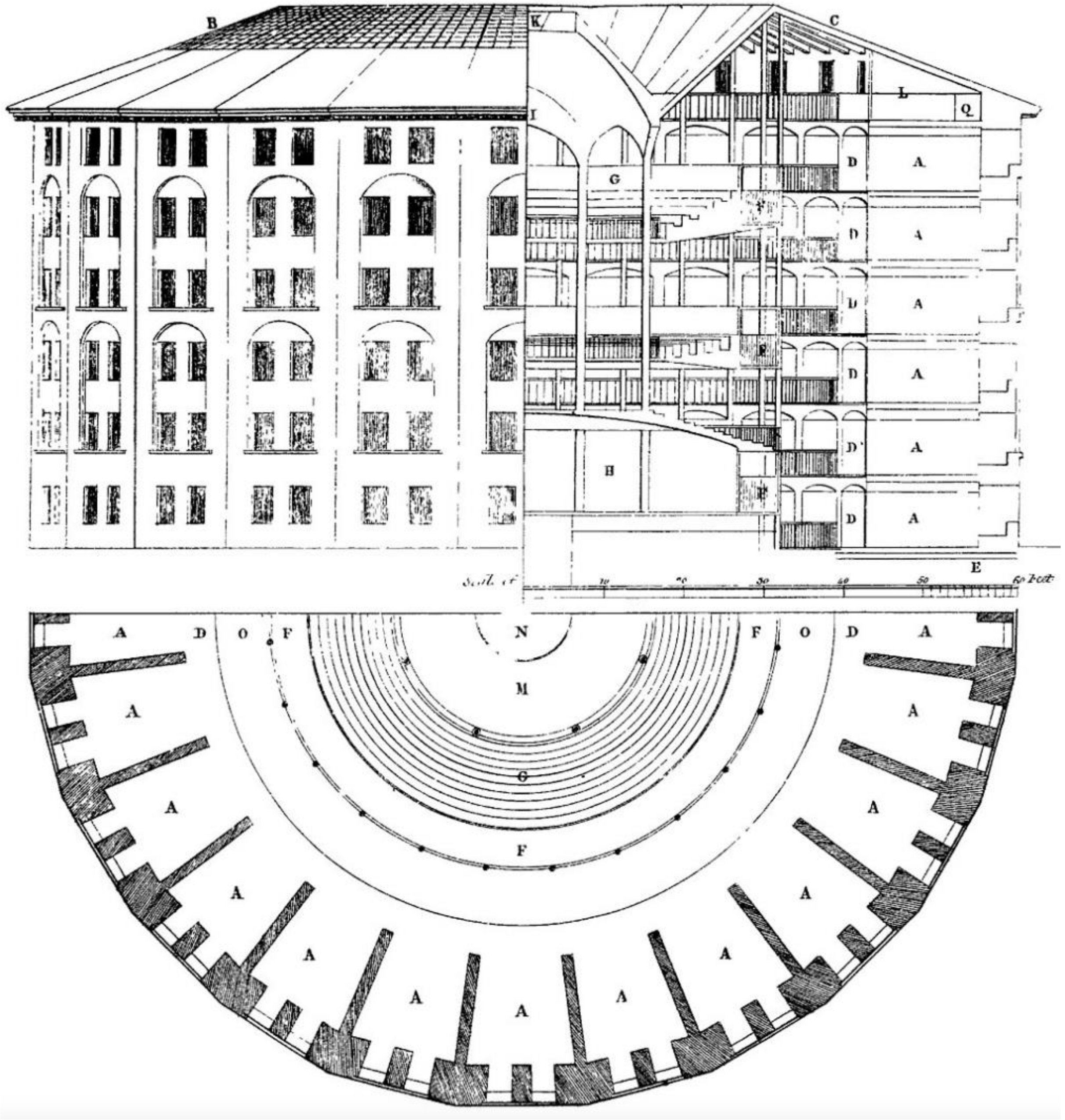
#BeFAIRandCARE

Microsoft's Responsible AI



https://github.com/microsoft/Data-Science-For-Beginners/tree/main/1-Introduction/02-ethics

Closing Thoughts



Jeremy Bentham


Special Issue 5: Grappling With the Generative A Published on Dec 30, 2024 DOI 10.1162/99608f92.21e6bbaa

Beware the Intention Economy: Collection and Commodification of Intent via Large Language Models

by Yaqub Chaudhary and Jonnie Penn

https://hdsr.mitpress.mit.edu/pub/ujvharkk/release/1

Ai = Data x (Representation $\Longrightarrow$ Loss) x Optimization

BIAS

Calibration

misalignment